



Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division

Digital Learning Handout

Part A: Content Design

Course Title	Software Engineering for Machine Learning
Course No(s)	AIMLCZG546
Credit Units	4
Credit Model	3-0-1
Course Author	Prof. Shreyas Rao
Version No:	1.0
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Course Description:

Machine Learning in Production: Introduction, From models to systems, Machine learning for software engineers; Requirements Engineering: When to use ML, Gathering requirements; Architecture and Design: Thinking like a software architect, Quality attributes of ML components, Automating ML pipeline, Scaling the system, Planning for operations; Coding practices: What is good code, Analyzing code performance, Using data structures effectively, OOP and functional programming, Errors, Logging & Debugging, Code formatting and linting, documentation, APIs; Sharing code: Version control, Dependencies, Packaging; Testing and Quality Assurance: Types of tests, Testing for ML, Testing model training, Testing model inference, Model quality, Data quality, Pipeline quality, System quality, Testing and experimenting in production; Automation and Deployment; Security for Machine Learning; Process and Teams: Data science and software engineering process models, Interdisciplinary teams; Responsible ML Engineering: Responsible Engineering, Provenance, Reproducibility, Interpretability and Explainability, Security and Privacy.

Course Objectives

No	Course Objective
CO1	Understand the transition from ML models to production-ready software systems
CO2	Apply software engineering principles to design and build ML-enabled systems
CO3	Analyze and define requirements, architecture, and quality attributes for ML applications
CO4	Develop and test robust, scalable ML pipelines using best coding and MLOps practices
CO5	Evaluate ethical, security, and operational aspects of ML systems in real-world deployment

Text Books

T1	"Machine Learning in Production: From Models to Products", by Christian Kästner, Carnegie Mellon University, MIT Press, April 2025
T2	"Software Engineering for Data Scientists" by Catherine Nelson, O'Reilly Publishers, April 2024



Reference Book(s) & other resources:

R1	“Moving from SDLC to ADLC” – White Paper from Tech Mahindra https://www.techmahindra.com/insights/whitepapers/moving-sdlc-ai-driven-software-development-lifecycle-adlc-generate-value/
R2	“A first look at the integration of machine learning models in complex autonomous driving systems: a case study on Apollo” Research Article, 2020 https://dl.acm.org/doi/10.1145/3368089.3417063
R3	“Software Engineering for Machine Learning: A Case Study”, Microsoft, 2019 https://www.microsoft.com/en-us/research/wp-content/uploads/2019/03/amershi-icse-2019_Software_Engineering_for_Machine_Learning.pdf
R4	"Generative to Agentic AI: Survey, Conceptualization, and Challenges", J.Schneider, arXiv:2504.18875v1, April 2025.
R5	"Cloud Native Artificial Intelligence", CNCF AI Working Group, 1st Edition, 2024
R6	GR4ML: https://www.cs.toronto.edu/~soroosh/gr4ml_introduction.html

Learning Outcomes: Students will be able to

LO1	Explain the system-level view of ML and its role within software products
LO2	Identify appropriate use cases and formulate requirements for ML-based solutions
LO3	Design scalable architectures and automated ML pipelines
LO4	Implement and evaluate ML systems using testing, versioning, and deployment practices
LO5	Assess ML systems in production considering performance, data quality, reliability, and ethical concerns

Modular Content Structure

M1: Foundations of ML Systems Engineering

- Introduction
- Software Engineering - Process Models, Roles
- Data science - Data Science Pipeline, Roles
- Machine Learning - Basic terminologies, ML Pipeline, Foundation Models, Types of ML Domains
- Machine Learning in production: Scope and Challenges
- From Models to Systems
- ML and non-ML components in a system
- AI Paradigms: Predictive, Generative, and Agentic AI
- Overview of Cloud-native ML Systems
- Case Studies

M2: Requirements Engineering for ML Systems

- Requirements Engineering
- When to use Machine Learning: ML as Predictions
- Setting and Measuring Goals
- From Goals to Requirements





- GR4ML Modeling Notation
- Defining, Composing and Evaluating Measures
- Quality Attributes of ML Components

M3: Architecture & Design

- Thinking like a software architect
- Common Architectural Design Challenges for ML-Enabled Systems
- Common System Architectures
- Architectural Patterns for ML Systems
- Microservices Pattern for ML
- Heartbeat Tactic
- Design Patterns for ML Systems
- Feature Store pattern
- RAG Model for LLMs

M4: Implementation and Code sharing

- Coding practices
- What is good code
- Analyzing code performance
- Using data structures effectively
- OOP and functional programming
- Errors, Logging & Debugging
- Code formatting and linting
- Research code vs production code
- Design and Refactoring
- Designing APIs for ML services
- Sharing code - Version control, Dependencies and Packaging

M5: Quality Assurance

- Types of tests
- Testing for ML
- Testing model training
- Testing model inference
- Model Quality
- Data Quality
- Pipeline Quality
- System Quality
- Testing and Experimentation in Production
- Security for Machine Learning

M6: Deployment

- Basics of Docker & Kubernetes
- Deployment strategies (batch, real-time, edge)
- Deploying model
- Model Inference Function
- Feature Encoding & Feature Stores





- Model Serving Infrastructure
- Types of Deployment models
- Documenting using Model Card
- Automating the ML Pipeline
- Stages of ML Pipeline
- Automation and Infrastructure Design
- Code Quality and Observability

M7: Responsible ML Engineering

- Responsible Engineering
- Versioning, Provenance and Reproducibility
- Explainability
- Fairness
- Safety
- Security and Privacy

M8: Application of SE Principles for Agentic AI

Part B: Learning Plan

Contact Session	List of Topic Title	Sub-Topics	Reference
1	Foundations of ML Systems Engineering	• Introduction • Software Engineering - Process Models, Roles • Data science - Data Science Pipeline, Roles • Machine Learning - Basic terminologies, ML Pipeline, Foundation Models, Types of ML Domains	T1-Chapter 1 T1-Chapter 3 Lecture Notes R1
2	Foundations of ML Systems Engineering	• Machine Learning in production: Scope and Challenges • From Models to Systems • ML and non-ML components in a system • AI Paradigms: Predictive, Generative, and Agentic AI • Overview of Cloud-native ML Systems • Case Studies	T1-Chapter 1 T1-Chapter 2 Lecture Notes R2, R3, R4, R5
3	Requirements Engineering for ML Systems	• Requirements Engineering • When to use Machine Learning: ML as Predictions • Setting and Measuring Goals • From Goals to Requirements • GR4ML Modeling Notation	T1-Chapter 4 T1-Chapter 5 T1-Chapter 6 T1-Chapter 7 Lecture Notes R6
4	Requirements Engineering for ML Systems Architecture &	• Defining, Composing and Evaluating Measures • Quality Attributes of ML Components • Thinking like a software architect	T1-Chapter 8 T1-Chapter 9 Lecture Notes





	Design		
5	Architecture & Design	- Common Architectural Design Challenges for ML-Enabled Systems - Common System Architectures - Architectural Patterns for ML Systems - Microservices Pattern for ML - Heartbeat Tactic	T1-Chapter 8 Lecture Notes
6	Architecture & Design	- Design Patterns for ML Systems - Feature Store pattern - RAG Model for LLMs	T1-Chapter 8 Lecture Notes
7	Implementation and Code sharing	- Coding practices - What is good code - Analyzing code performance - Using data structures effectively	T2-Chapter 1 T2-Chapter 2 T2-Chapter 3 Lecture Notes
8	Implementation and Code sharing	- OOP and functional programming - Errors, Logging & Debugging - Code formatting and linting - Research code vs production code	T2-Chapter 4 T2-Chapter 5 T2-Chapter 6 Lecture Notes
9	Implementation and Code sharing	- Design and Refactoring - Designing APIs for ML services - Sharing code - Version control, Dependencies and Packaging	T2-Chapter 8 T2-Chapter 10 T2-Chapter 11 Lecture Notes
10	Quality Assurance	- Types of tests - Testing for ML - Testing model training - Testing model inference - Model Quality - Data Quality	T1-Chapter 14 T1-Chapter 15 T1-Chapter 16 Lecture Notes
11	Quality Assurance	- Pipeline Quality - System Quality - Testing and Experimentation in Production - Security for Machine Learning	T1-Chapter 17 T1-Chapter 18 T1-Chapter 19 Lecture Notes
12	Deployment	- Basics of Docker & Kubernetes - Deployment strategies (batch, real-time, edge) - Deploying model - Model Inference Function - Feature Encoding & Feature Stores - Model Serving Infrastructure	T1-Chapter 10 Lecture Notes
13	Deployment	- Types of Deployment models - Documenting using Model Card - Stages of ML Pipeline - Automating the ML Pipeline - Automation and Infrastructure Design - Code Quality and Observability	T1: Chapter 11 Lecture Notes





14	Responsible ML Engineering	- Responsible Engineering - Versioning, Provenance and Reproducibility - Explainability - Fairness - Safety - Security and Privacy	T1: Chapters 23-29
15	Application of SE Principles for Agentic AI	- Phases of AI-Driven Software Development Life Cycle (ADLC) - Application of SE Principles for Agentic AI	Lecture Notes
16	Application of SE Principles for Agentic AI & Course Review	- Important Agentic AI patterns - Course Wrap-up and Review	Lecture Notes

Experiential Learning Components:

1. Lab work: 6
2. Project work: 0
3. Case Study: 2
4. Simulation: 0
5. Work Integrated Learning Assignment- 2 Assignments
6. Design work/ Field work: 0

Objective of Experiential Learning Component:

Hands on sessions on implementation of:

- a) End-to-end ML system development, covering problem formulation, requirements, and system design
- b) Design and development of scalable ML architectures and modular pipelines with versioning and experiment tracking
- c) Testing, validation, and quality assurance of ML systems across data, model, and pipeline levels
- d) Automation and deployment of ML pipelines using containerization and orchestration frameworks

Scope of Experiential Learning Component:

Programming Language: Python

Tools and Platforms:

Pipeline orchestration: Prefect, Prefect Cloud

Experiment tracking and model management: MLflow

Data versioning: DVC

Cloud ML platform: Amazon SageMaker

Containerization and orchestration: Docker, Kubernetes (Minikube)

API development: FastAPI

Testing and monitoring: PyTest, Evidently AI

Lab Infrastructure:

Online/ Open source/ Virtual Lab/ Google Colab



List of Experiments:

Lab No	Lab Objective	Session Reference
1	Create an end-to-end ML system blueprint for a real-world use case (e.g., fraud detection or recommendation system)	3
2	Formulate requirement specifications and define measurable goals for an ML-based application using GR4ML concepts	4
3	Design a scalable ML system architecture using microservices and feature store patterns	6
4	Design and develop a modular ML pipeline with versioning, packaging, and experiment tracking	9
5	Design and develop a testing and validation framework for ML pipelines covering data, model, and system quality	11
6	Design and develop an automated ML pipeline with deployment using containerization and orchestration	13

Evaluation Scheme:

Legend: EC = Evaluation Component; AN = After Noon Session; FN = Fore Noon Session

Evaluation Component	Name (Quiz, Lab, Project, Mid-term exam, End semester exam, etc.)	Type (Open book, Closed book, Online, etc.)	Weight	Duration	Day, Date, Session, Time
EC – 1	Quiz	Online	05%	3 days	To be announced
	Situated Learning	Online	05%	3 days	To be announced
	Assignment I & II	Online	20%	20-30 days	To be announced
EC - 2	Mid-Semester Test	Closed Book	30%	2 hours	To be announced
EC - 3	Comprehensive Exam	Open Book	40%	2 ½ Hours	To be announced

Syllabus for Mid-Semester Test (Closed Book): Topics in Contact session: 1 to 8

Syllabus for Comprehensive Exam (Open Book): All topics

Important Links and Information:

eLearn Portal: <https://elearn.bits-pilani.ac.in>

Students must visit the eLearn portal regularly and stay updated with the latest announcements and deadlines.



Contact Sessions: Students should attend the online lectures as per the schedule provided on the eLearn portal.

Evaluation Guidelines:

1. EC-1 consists of either two Assignments or three Quizzes. Students will attempt them through the course pages on the eLearn portal. Announcements will be made on the portal in a timely manner.
2. For Closed Book tests: No books or reference material of any kind will be permitted.
3. For Open Book exams: “open book” means text/ reference books (publisher copy only) and does not include any other learning material. No other learning material will be permitted during the open book examinations. For Detailed Guidelines refer to the attached document.

[EC3 Guidelines](#)

4. If a student is unable to appear for the Regular Test/Exam due to genuine exigencies, the student should follow the procedure to apply for the Make-Up Test/Exam, which will be made available on the eLearn portal. The Make-Up Test/Exam will be conducted only at selected exam centres on the dates to be announced later.

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course handout, attend the online lectures, and take all the prescribed evaluation components such as Assignments/Quizzes, Mid-Semester Tests and Comprehensive Exams according to the evaluation scheme provided in the handout.

